

Important Notes:

1. First, check the problem sheets for completeness:
 - 1 title page
 - 1 page with exam instructions
 - 2 blank pages
 - 12 problem pagesThis makes a total of 16 pages.
2. Write the solutions directly after the corresponding problem on the respective pages. If you find during the process that there is not enough space on the problem sheets, use the two blank pages immediately following the instructions or, if necessary, the back side. Write the solutions with full specification of the subtask you are addressing (e.g., "Task 2a").
3. If the available space is still insufficient, the examiner or a supervisor will provide additional paper.
4. The cover sheet, as well as all problem pages or other pages related to the problems, must be labeled with your matriculation or seat number (for VIN).
5. Problems or subproblems are considered solved only if the individual solution steps are legible and understandable. Only one solution per problem may be submitted.
6. The metal clip holding the papers together must not be removed.

Task 1: Basic Concepts

- a) Describe the relationship between the terms *artificial intelligence*, *machine learning*, and *deep learning*, and distinguish them from one another. How do classical machine-learning methods and deep-learning methods differ with respect to feature extraction? **(4 P)**
- b) What is the difference between supervised and unsupervised learning? Which of these approaches is typically used for classification problems? **(3P)**
- c) What is meant by the *train/test split*, and what is it typically used for in the context of supervised learning? **(2P)**

- d) Compare the two phenomena *overfitting* and *underfitting*. Fill in the following table. Use only the terms “high” and “low.” (3P)

Criteria	Overfitting	Underfitting
Amount of Error on the train data		
Amount of Error on the test data		
Model Complexity		

- e) A confusion matrix is given for evaluating a binary classifier that labels emails as *spam* or *not spam*. Use it to calculate the metrics *precision* and *recall*. What does “true positives” mean? (5 P)

		True Class	
		Spam	Kein Spam
Prediction	Spam	True Positives 10	False Positives 20
	Kein Spam	False Negatives 30	True Negatives 10

f) Below are short descriptions of specific AI problem settings. For each one, decide whether it is a classification, regression, or clustering problem. Justify your decision. **(4 P)**

- **Flight delay:** Based on historical flight data, the goal is to estimate by how many minutes a particular flight will be delayed.
- **Medical diagnosis:** Using medical measurements (e.g., body temperature) and symptoms, the goal is to predict whether a particular disease is present.

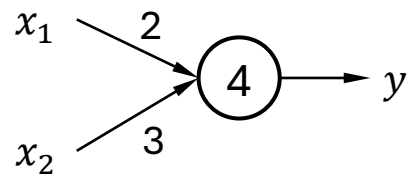
g) Briefly explain the basic principle of the *gradient descent* method. What effect does choosing a learning rate that is too high have in this method? **(4 P)**

h) Why is the *backpropagation* algorithm a computationally efficient method for calculating gradients in multi-layer neural networks? **(2 P)**

Task 2: Threshold Logic Units (TLUs)

a) Describe the general structure of a Threshold Logic Unit. **(3 P)**

b) Which Boolean function is modeled by the following TLU? Justify your decision. **(2 P)**



c) Which of the following Boolean functions can be modeled perfectly by a single threshold logic unit, so that every input combination produces the correct output? Justify your decision.

Given Boolean functions: AND, Implication, Bi-implication. **(3 P)**

- d) Apply the *delta rule* using the training data provided below to the given threshold logic unit. Perform only a single training iteration. Are further iterations required?
(8 P)

Training Data:

$$S = \{(x_1 = 1, x_2 = 2, y = 0)\}$$

Initialisation of the Threshold Logic Unit:

$$w_0 = 1, w_1 = 3, w_2 = 1$$

(w_0 represents the weight of a constant input 1 and models the trainable threshold)

Learn rate:

$$\eta = 0,5$$

Task 3: Multilayer Perceptrons (MLPs)

- a) What is the essential difference between networks of threshold logic units and multilayer perceptrons? What relevance does this difference have for the gradient descent method? **(2 P)**

- b) The following Python code shows the implementation of a simple multilayer perceptron model. However, errors occur when running the code. Analyze and correct the code if necessary in subpoints i to v. **(total 12 P)**

Python-Code:

```
1  import numpy as np
2  import tensorflow as tf
3
4  X = np.array([
5      [1.0, 2.0, 0.5, 0.3], # → Class: Cat
6      [3.0, 1.0, 0.2, 0.1], # → Class: Dog
7      [0.1, 0.4, 2.5, 2.0], # → Class: Hare
8      [2.0, 1.5, 0.4, 0.3], # → Class: Dog
9      [0.2, 0.3, 2.0, 1.8], # → Class: Hare
10 ])
11
12 y = np.array([
13     [1, 0, 0], # Cat
14     [0, 1, 0], # Dog
15     [0, 0, 1], # Hare
16     [0, 1, 0], # Dog
17 ])
18
19 model = tf.keras.Sequential([
20     tf.keras.layers.Input(shape=(8,)),
21     tf.keras.layers.Dense(units=4, activation='relu'),
22     tf.keras.layers.Dense(units=2, activation='relu'),
23     tf.keras.layers.Dense(units=1, activation='relu')
24 ])
25
26 model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
27
28 model.fit(X, y, epochs=50, batch_size=16)
```

- i. What do the variables X and y mean? **(2 P)**

-
- ii. Why does the combination of the values for the two variables X and y cause an error during the training process? What would the corresponding correction look like? **(2 P)**

 - iii. How many trainable weights does the multilayer perceptron have? Assume that the MLP uses bias neurons. **(2 P)**

 - iv. The model implementation in lines 19 to 24 contains errors. Identify these errors and provide a possible correction for each. **(4 P)**

 - v. Why does line 28 cause an error during code execution? Provide a possible correction. **(2 P)**

Task 4: Convolutional Neural Networks (CNNs)

- a) What types of layers does a CNN typically consist of when used as a classifier, e.g., for images? Which of these layers are typically used for feature extraction? Why is it useful for certain types of layers to appear multiple times in the model? **(6 P)**
- b) Calculate the result of applying a filter K (i.e., the result of a convolution) to the input I with padding $P = 1$ and stride $s = 3$. Compute the corresponding entries of the resulting feature map using the ReLU activation function. (ReLU: $f(x) = \max(0, x)$). **(9 P)**

$$I = \begin{bmatrix} 5 & 2 & 1 \\ 3 & 8 & 6 \\ 4 & 7 & 2 \end{bmatrix}, K = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

- c) What is the result of the max-pooling operation at the specified position of the feature map (position marked with a dashed frame)? Describe the basic principle of max-pooling in one sentence. **(2 P)**

1	-4	5	7
2	0	1	1
3	9	9	9
0	0	-2	0

- d) Why is it not recommended to use multilayer perceptron models for image classification? Name two reasons. **(2 P)**

Task 5: Recurrent Neural Networks (RNNs)

- a) What difference compared to MLPs allows RNNs to process sequential data?
Express the processing principle as a recursive function. **(3 P)**
- b) What is generally meant by the vanishing gradient problem? Which specific property of Elman networks (as a type of RNN) leads to the vanishing gradient problem?
Name one other general cause. **(4 P)**
- c) Consider an Elman RNN with input size 10, one hidden layer of size 20, and output size 5. Compute the number of trainable parameters in the RNN, do **not** include bias neurons for both the hidden and output layers. **(2 P)**

Task 6: Reinforcement Learning

- a) Explain the concept of reinforcement learning by explaining the relations between agents, the environment and rewards. **(3 P)**

- b) Explain each underlined component of the Bellman equation that is given by

(4 P)
$$Q_{new}(s, a) = \underline{Q(s, a)} + \alpha [\underline{R(s, a)} + \underline{\gamma} \cdot \underline{\max Q'(s', a')} - \underline{Q(s, a)}]$$

- c) You are given a simple reinforcement learning setting with Q-learning. Consider an agent interacting with a deterministic environment with two states and two actions. Describe the algorithm and fill in the missing values in the Q-table representations by calculating the new Q-values after each step. **(8 P)**

Environment: States: $S = \{s_1, s_2\}$, Actions: $A = \{a_1, a_2\}$, Learning rate: $\alpha = 1$, Discount factor: $\gamma = 0.5$, Rewards r

Experienced sequence: Step 1: $s_1 \xrightarrow{a_1, r=2} s_2$, Step 2: $s_2 \xrightarrow{a_1, r=4} s_1$

Use the Bellman equation given in task 6b.

Algorithm description:

Step 1:

State/Actions	a_1	a_2
s_1		
s_2		

Step 2:

State/Actions	a_1	a_2
s_1		
s_2		